Calcurion Physics Worksheets

A-Level Physics

Topic: Circular Motion & SHM

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define angular speed ω. **[1]**

2. A particle moves in a circle of radius 0.50 m at 2.0 rev/s. Calculate ω and linear speed v. **[3]**

3. State the equation for centripetal acceleration. **[1]**

4. Calculate the centripetal force on a 0.20 kg mass moving at 3.0 m/s in a circle of radius 0.40 m. **[3]**

5. What provides the centripetal force for a car on a flat circular track? **[1]**

6. Define simple harmonic motion. **[2]**

7. State the condition for SHM: $a = -\omega^2 x$. **[1]**

8. A mass on a spring has $T = 0.50$ s. Calculate ω. **[2]**

9. State the equation for the period of a mass–spring system. **[1]**

10. A 0.25 kg mass on a spring of $k = 100$ N/m oscillates. Calculate the period. **[3]**

11. State the equation for the period of a simple pendulum. **[1]**

12. A pendulum of length 1.0 m has period? Take $g = 9.8$ m/s^2. **[2]**

13. Define amplitude. **[1]**

14. At what position in SHM is the speed maximum? **[1]**

15. At what position in SHM is the acceleration maximum? **[1]**

Section B – Hard (Q16–25)

16. A satellite orbits Earth at radius 7.0×10^6 m. Calculate its orbital speed. ($M_E = 6.0 \times 10^{24}$ kg, $G = 6.67 \times 10^{-11}$) **[3]**

17. Derive $T = 2\pi\sqrt{l/g}$ for a simple pendulum from $a = -\omega^2 x$. **[4]**

18. A mass–spring system has amplitude 5.0 cm and period 0.40 s. Calculate maximum speed and maximum acceleration. **[4]**

19. Sketch graphs of displacement, velocity and acceleration against time for SHM, showing phase relationships. **[4]**

20. Energy in SHM is $\frac{1}{2}kA^2$. Show that total energy is constant. **[3]**

21. A particle in SHM has $x = 0.04\cos(5\pi t)$. State amplitude, ω and frequency. **[3]**

22. Calculate the maximum KE of a 0.10 kg mass oscillating with amplitude 8.0 cm and period 0.60 s. **[3]**

23. Explain why a pendulum clock runs slow if taken to a mountain (higher altitude). **[2]**

24. A car takes a banked curve of radius 50 m at 20 m/s. Calculate the ideal banking angle. ($g = 9.8$) **[3]**

25. Describe an experiment to investigate the relationship between period and mass for a mass–spring system. **[3]**

End of Worksheet • *Show all working clearly.*